Vatsal Verma

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PROFESSIONAL SUMMARY

Software Engineer with 3 years of experience developing scalable Java and Spring Boot microservices and building event-driven systems using Apache Kafka. Proficient in designing and implementing REST APIs, building efficient data access layers in PostgreSQL and Couchbase, and managing CI/CD pipelines on Kubernetes and Red Hat OpenShift. Over the past year, expanded into Agentic AI, building AI agents, RAG systems, and custom MCP servers in Java and Spring to integrate agents with production data and backend services for faster incident diagnosis and automated reprocessing.

TECHNICAL SKILLS

Languages: Java, Python, SQL

Backend & APIs: Spring Boot, Spring Cloud, Spring Security, Microservices, REST APIs, OpenAPI, OAuth2, JWT, API Rate Limiting

Data & Messaging: Apache Kafka, Couchbase, PostgreSQL, Elasticsearch, Event-Driven Architecture

AI / GenAI: Agentic AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), LangGraph, Google Agent Development Kit (ADK), Model Context Protocol (MCP), Deep Java Library (DJL), Embeddings, Semantic Search, BM25, kNN, Hybrid Retrieval

Cloud & DevOps: AWS, Docker, Kubernetes, Red Hat OpenShift, Jenkins, Maven, CI/CD, Linux, Git

Testing & Quality: JUnit, TestNG, Mockito, SonarQube

PROFESSIONAL EXPERIENCE

AI Software Engineer *(SWE-2)* *August 2025 – Present*
Amdocs, Pune, India

- Engineered a production-grade Agentic AI system with LangGraph and Google ADK that retrieves context via hybrid RAG (BM25 + kNN) over 40k+ indexed RCA and ontology chunks in Elasticsearch, achieving 500ms p95 retrieval latency and reducing incident diagnosis time by 20%.
- Designed and developed a Spring Boot RAG ingestion pipeline that processes ontology and RCA documents through Apache Kafka, chunks the content, generates embeddings with Deep Java Library (DJL), and indexes the data in Elasticsearch for retrieval.
- Implemented Java MCP servers using the Streamable HTTP transport with tools to query microservice logs in Elasticsearch and fetch business-specific documents from Couchbase, so the agent can draft a first-pass RCA from live production data.

Software Developer *(SWE-1)* *August 2023 – July 2025*
Amdocs, Pune, India

- Owned two Java, Spring Boot, and Spring Cloud microservices at a time on a rotating basis for Real-Time Billing (RTB), a 36-service B2B/B2C billing product used by AT&T and Bell Canada that serves 80M+ subscribers and processes 10M+ billing events daily.
- Implemented REST APIs (OpenAPI), secured them with Spring Security (OAuth2/JWT), applied API rate limiting, produced and consumed billing events on Apache Kafka, and used Caffeine on the request path as cache-aside for frequent KV reads from Couchbase buckets handling 50k+ ops/sec at sub-10ms latency.
- Optimized CI/CD and integration testing on Red Hat OpenShift by using Kubernetes APIs to run Maven-based TestNG suites across multiple clusters, reducing test execution time by 25%.
- Resolved 42 CVEs in direct and transitive Maven dependencies on the runtime classpath, and replaced string-concatenated N1QL with parameterized Couchbase queries to prevent injection (OWASP A05:2025).

AI/ML Research Intern *August 2021 – February 2022*
Indian Institute of Technology (IIT-BHU), Varanasi, India

- Developed human facial-recognition models using Convolutional Neural Networks (CNNs) and Histogram of Oriented Gradients (HOG) in TensorFlow and Keras. Trained models on a 50,000-image dataset through computer vision and image-processing pipelines.
- Adapted the facial-recognition pipeline to livestock identification using muzzle patterns across 100+ cattle, then evaluated the models and presented findings to faculty advisors.

PROJECTS

Access Control for Multi-Agent MCP Servers (2026) *[Spring Boot, Spring Security, MCP]* *github.com/vatsal259/ArwenForge*

- Built an open-source Java 21 library that enforces profile-based RBAC on multi-agent MCP servers so agents only discover and invoke allowed tools, resources, and prompts.
- Developed Spring Boot auto-configuration, Spring Security (JWT and API-key authentication), signed session tokens for agent delegation, and a CLI for policy validation and blast-radius reports.
- Implemented approval gates that block sensitive MCP tool calls until granted, and instrumented that path with allow/deny audit events, per-agent rate limits, and Micrometer metrics.

File-Based Fuzzy Search Library (2026) *[Java 21, Maven, Trie Indexing, Levenshtein]* *github.com/vatsal259/docube-db*

- Built a file-based fuzzy-search open-source library that indexes codes into a chunked trie, pruning candidates by shared prefix to avoid full-catalog scans, enabling offline lookup over 50k+ invoice, shipment, and serial IDs.
- Implemented four recall strategies (exact, prefix, single-typo walk, beam search) and ranked survivors with Levenshtein scoring to recover OCR and scanner typos without a database or server.
- Designed a single Maven module with Jackson as the only runtime dependency, persisting indexes on disk for offline, laptop-scale lookup over large code lists.

EDUCATION

Bachelor of Technology in Computer Science and Engineering with Specialization in IoT *July 2019 – May 2023*
Vellore Institute of Technology, Vellore, India
CGPA: 8.22/10

ACHIEVEMENTS AND LEADERSHIP

- **Innovation Award**, Amdocs, 2025, awarded for CI/CD and integration-test optimization reducing execution time by 25%.
- **Most Creative Use of the Twilio API**, WildHacks 2021, built a real-time SMS notification system for emergency alerts.
- **Design Lead**, ISA-VIT Student Chapter (2021–2022), led a 15-person team to design and ship Spark, an Android app, to the Google Play Store.